

Token Provisioning Best Practices Webinar



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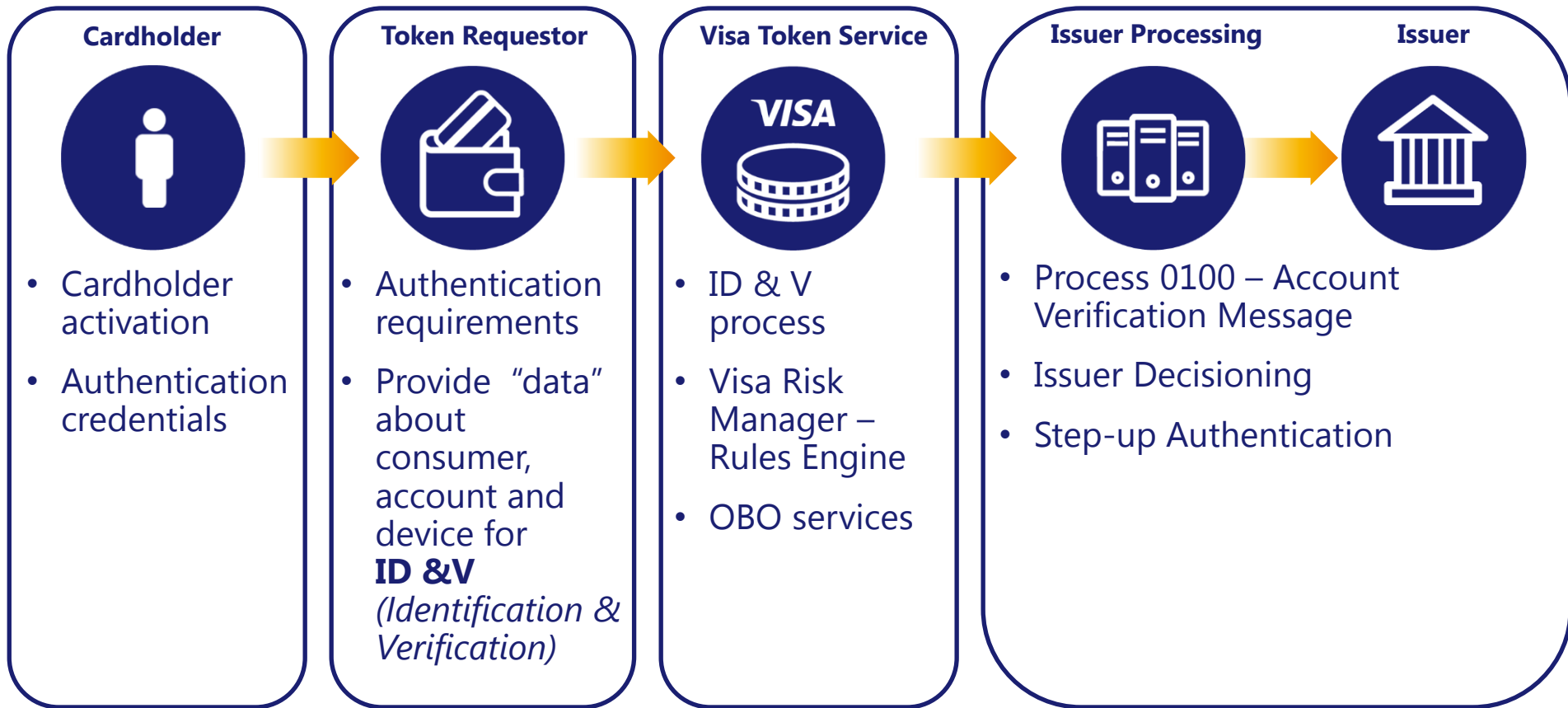


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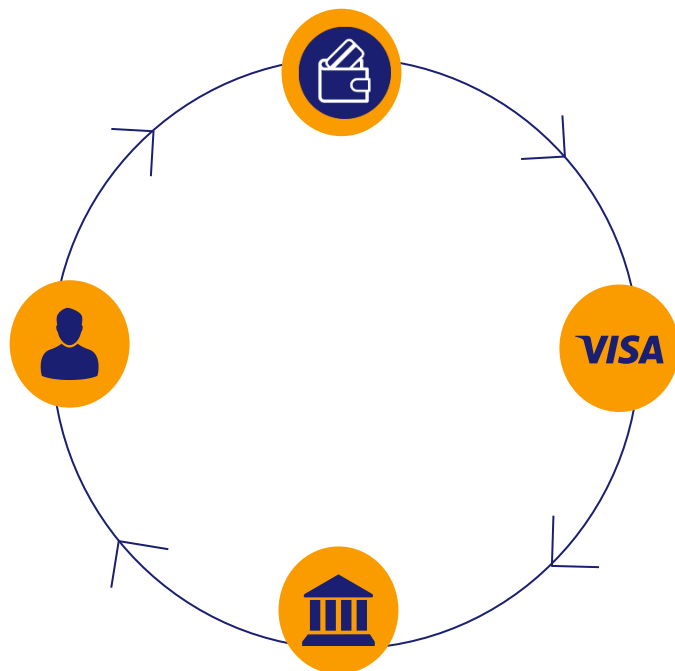
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Digital Solutions: Overview of Token Provisioning Process

Overview of Token Provisioning



Identification & Verification



- ✓ Fraud Review
- ✓ Issuer managed authentication
- ✓ Best practices

Visa Token Service – Token Provisioning



Additional authentication requirements

Using the Visa Token Service, issuers set up criteria that allows Visa to **approve**, token requests or refer to **manual step-up** on issuer's behalf.

The ID & V process during activation / provisioning may result in the digital solution requesting:

- Issuers requiring to perform additional authentication (step up)
- Card Account and Device cannot be activated for payments without additional authentication by issuer

Examples of conditions* that may require additional authentication by issuer:

- Customer ID/Card pair is newer than date threshold
- Customer ID too new relative to launch; relating to provisioning request
- Changes made recently to account data
- Suspicious transactions linked to customer's existing account
- Account has had no activity in a year

Issuer Decisioning Options



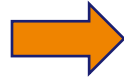
1. Use Visa Risk Manager to setup criteria to “approve,” or “manual step up” token requests
 - Visa will apply issuer criteria during token provisioning request
2. Prepare/Expect to receive standard ISO 0100 Account Verification message for the card account in each request (business as usual)
 - Request will contain PAN, EXP DT, CVV2 (based on issuer option) and address data (when present)
 - *Note: No token related data is passed in this message*
3. Respond with an account status check and Address Verification result code as applicable
4. Set up call center for cardholders to call and provide details to complete the additional authentication requirements. (resolution option)
5. Enhance Mobile Banking App as means to complete step up authentication

Issuer Resolution Options

Issuers options to handle the cardholder enhanced authentication requirement.

1

Authenticate Cardholder



Call Center setup

Use Call center to complete Authentication

- Issuer's own call center (or)
- Processor service (or)
- Visa Global Visa Customer Care Services (GCAS)



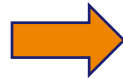
Mobile Banking App

Use mobile banking to complete Authentication

- Cardholder authenticates through mobile banking App

2

Activate Token for Payment



Use Token lifecycle tool through Visa Online

- Help desk users query PAN and activate Tokens



Use Batch Token Maintenance file to activate tokens

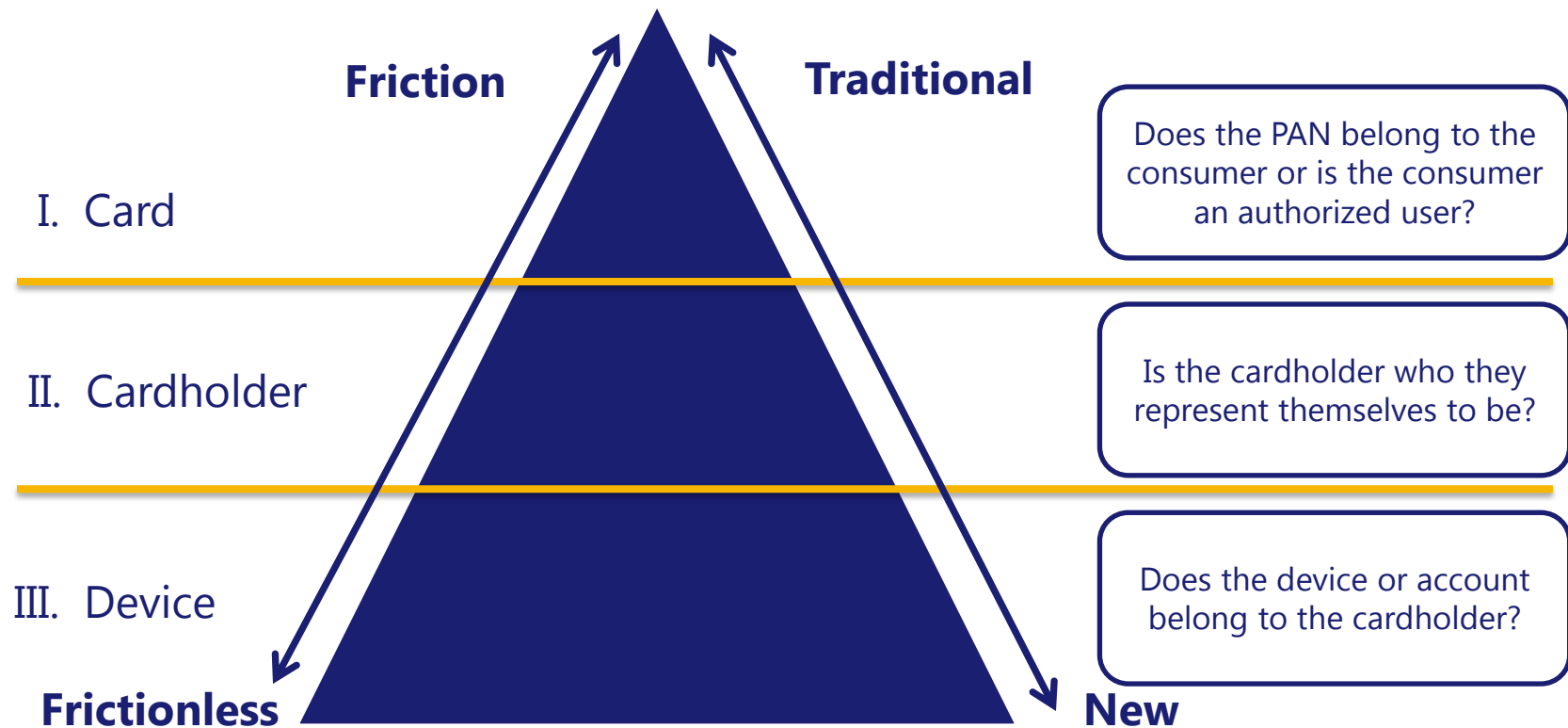
- Issuers / processors to submit file through OFD to activate tokens

Risk Best Practices: Token Provisioning

Identity and Verification



Triangulation of Card, Cardholder and Device create a unique digital identity through tokenization



Card & Cardholder Authentication



Card Authentication

1. CVV2 validation
2. Expiration date

Cardholder Authentication

1. AVS (Address Verification Service) Check
2. Digital solution should provide cardholder authentication functions
3. Confirm enrollment through additional channels *on file* with Issuers
4. OTP and online banking can be used to validate the cardholder. More difficult to social engineer than call centers. Preferred option
5. Call centers should validate cardholder assuming fraudsters have personally identifiable and card information available
 1. Recent banking transactions
 2. Recurring transactions
 3. Other expenses

1. Mobile phone geo-location (if applicable) should be matched with the address on file at the Issuer
2. Monitor token provisioning, de-provisioning and purchase velocity on a device to detect any suspicious behavior
3. Monitor zip code of other enrolled cards on the device during provisioning
4. Accounts provisioned with missing billing information and geo-location should be carefully monitored

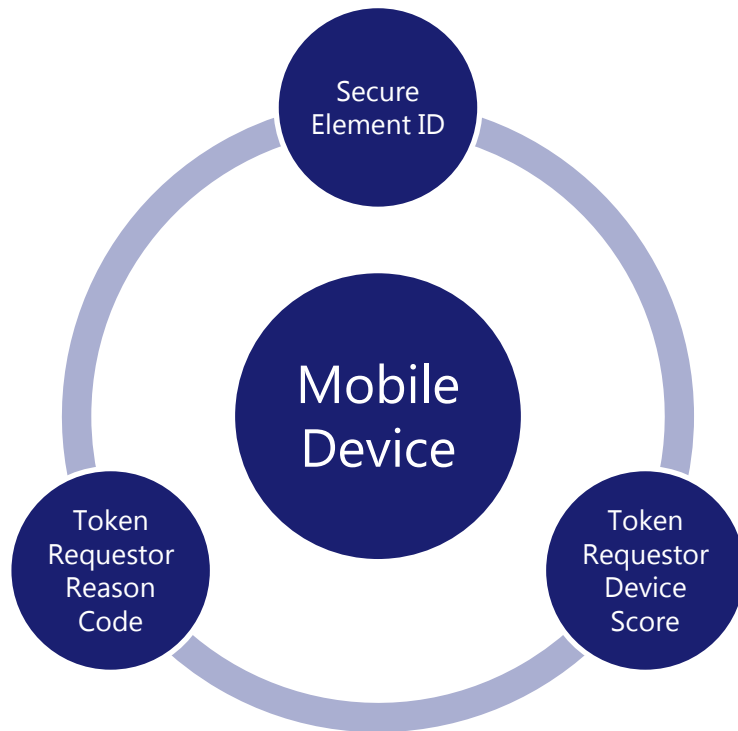
Transaction Processing



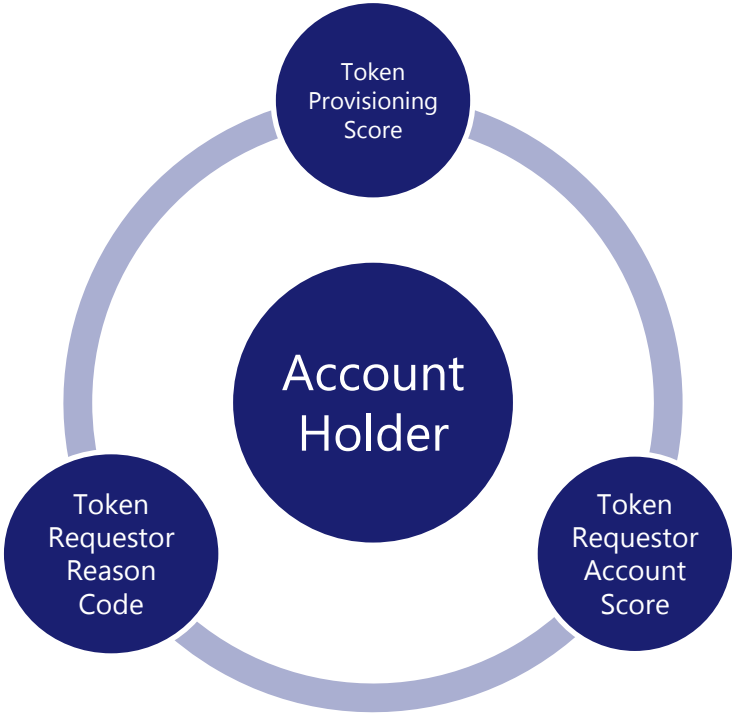
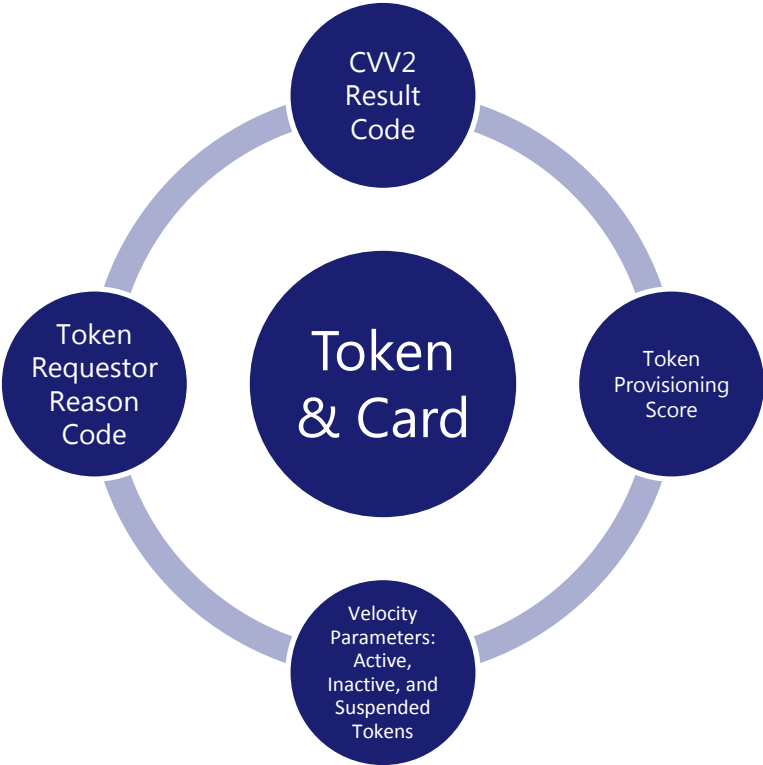
1. Closely monitor all transactions originating from token provisioned using step up authentication
2. Issuers can adjust transaction profile based on the Application Transaction Counter (ATC) which can measure age of the token

Visa Risk Manager Best Practices: Token Provisioning

Visa Risk Manager Token Parameters



Visa Risk Manager Token Parameters



Risk Management Tool – Suggested Rules



Rule Description	Action	Timeframe	Rule Criteria
Retry token requests when CVV2 mismatches	Retry	Always	CVV2 Result Code <> M
Remand risky token requests ¹	Manual Step-up	Always	Token Requestor Reason Code = Null AND Token Provisioning Score >= 90 ¹
Remand risky token requests ¹	Manual Step-up	Always	Token Requestor Reason Code = 06 AND Token Provisioning Score >= 85 ¹
Remand token requests from known suspect Secure Element IDs	Manual Step-up	Always	Secure Element ID IN (Blacklist)
Remand token requests that reach specific thresholds	Manual Step-up	Always	Number of Active Tokens for Account >= 3
Remand tokens requests with suspicious Token Requestor Reason Codes	Manual Step-up	Always	Token Requestor Reason Code = 0B

¹Based on an issuer's risk tolerance, issuers should select their threshold for the Token Provisioning Score using the Visa Token Provisioning Score Model Description in the Appendix of this document.

Risk Management Tool – Token Provisioning Score

Visa Token Provisioning Model Description

- Visa calculates the Token Provisioning Score using a custom model built for the Visa Network Token Service. The primary input to this initial model leverages account profiles built on historical spending patterns for each account.

For a more lenient approach where cardholder impact is minimized the following rules could be deployed where a token request is declined:

Token Provisioning Score	Token Requestor Reason Code ¹	Cumulative Transaction Volume Impacted	Cumulative Fraud Detected	False Positive Ration
>=90	Null	0.04%	2.4%	7.4:1
>=85	06	0.15%	8.9%	7.5:1

For a more aggressive approach where potential cardholder impact is increased while at the same time reducing more fraud, the following rules could be deployed where a token request is declined:

Token Provisioning Score	Token Requestor Reason Code ¹	Cumulative Transaction Volume Impacted	Cumulative Fraud Detected	False Positive Ration
>=75	Null	0.24%	6.7%	14.0:1
>=70	06	0.46%	14.9%	13.9:1

¹Token Requestor Reason Codes are indicators that there may be reason for suspicion for a token request. Null indicates that there isn't anything suspicious from the requestor's side, and 06 indicates that there has been no activity on this account for the past year.

Q&A

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